

Practical Chemistry Model Record & Viva Answers

Step-by-Step Observation Tables & Top 25 Viva Questions

Chapter / Focus Area: Practical Chemistry Solutions

Volumetric Calculations & Top Viva Voce Answers

1. Calculation Formula: $(M_1 \times V_1)/n_1 = (M_2 \times V_2)/n_2$

Where $n_1 = 2$ (for KMnO_4) and $n_2 = 10$ (for Mohr's Salt). Strength = Molarity * Molar Mass (158 g/mol).

2. Why is dil. H_2SO_4 used instead of HCl or HNO_3 in KMnO_4 titrations?

Answer: HCl is oxidized by KMnO_4 into Cl_2 gas, giving higher titre value. HNO_3 is itself a strong oxidizing agent. H_2SO_4 is stable and provides required acidic medium.

3. What is the chemistry of the Brown Ring Test for Nitrate?

Answer: Nitrate is reduced to NO by Fe^{2+} , which complexes with $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ to form brown nitroso ferrous sulphate $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$.

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Web: <https://pankaj2816.github.io/ajay-choudhary-chemistry/> • Contact: contact@ajaychemistry.com